



CHAPTER - 1

SOME BASIC CONCEPTS OF CHEMISTRY



1.1 Importance of Chemistry

- Chemistry is the study of matter, its composition, structure, properties and the changes it undergoes.
- It helps in understanding the substances around us and their reactions.
- It has vast applications in medicine, agriculture, industry, environment, materials, etc.
- Chemistry is often called the "Central Science" as it connects physics, biology and other natural sciences.

In Real Life

- ✓ Cooking food
- ✓ Making medicines
- ✓ Metals extraction
- ✓ Fuels & energy
- ✓ Plastic, glass, soap
- ✓ Dyes, paints, etc.

1.2 Nature of Matter

Matter is anything that has mass and occupies space.
Everything around us is made up of matter.

Characteristics of Particles of Matter

- * Matter is made up of extremely small particles.
- * These particles have spaces between them.
- * They are continuously moving.
- * They attract each other.
- * The kinetic energy of particles increases with rise in temperature.

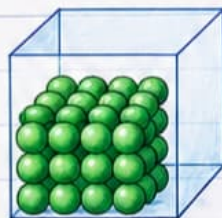
1.3 States of Matter

State	Characteristics	Example
Solid	• Definite shape & volume • Particles tightly packed • Very less compressible	Ice, stone, iron, sugar
Liquid	• Definite volume, no fixed shape • Particles close but mobile • Slightly compressible	Water, oil, milk
Gas	• No fixed shape or volume • Particles far apart • Highly compressible	O ₂ , CO ₂ , steam

Arrangement of particles in three states

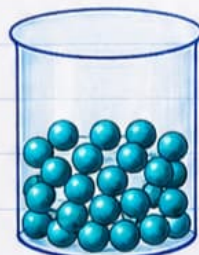
Solid

Particles are very close to each other.
(Fixed shape and volume)



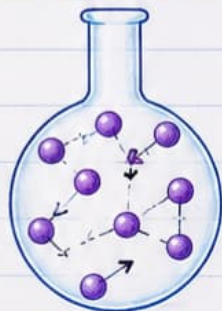
Liquid

Particles are close but can move over each other.
(Fixed volume, no fixed shape)



Gas

Particles are far apart and move freely in all directions.
(No fixed shape or volume)



Key Point

- * All matter is made of tiny particles.
- * These particles are always in motion.
- * Temperature affects the speed and energy of particles.

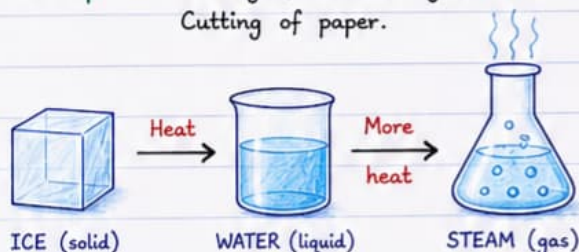


1.4 Physical and Chemical Changes

Physical Change

- Only physical properties like size, shape, state etc. change.
- No new substance is formed.
- Generally reversible.

Examples : Melting of ice, Boiling of water, Cutting of paper.



Chemical Change

- New substances with new properties are formed.
- Generally not reversible.
- Involves chemical reaction.

Examples : Rusting of iron, Burning of paper, Souring of milk.



Rusting of iron
(Iron + Oxygen + Moisture $\rightarrow$ Rust)

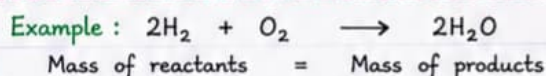


Burning of paper
(Paper + Oxygen $\rightarrow$ Ash + Gases)

1.5 Laws of Chemical Combination

* Law of Conservation of Mass (Lavoisier)

The total mass of reactants is always equal to the total mass of products.



Mass can neither be created nor destroyed in a chemical reaction.



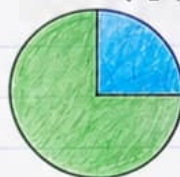
Mass is always conserved.

* Law of Definite Proportions (Proust)

A given compound always contains the same elements combined in a fixed proportion by mass.

Example : Water (H_2O) always contains Hydrogen and Oxygen in the ratio 1 : 8 by mass.

Water (H_2O)



Oxygen = 8 parts (88.9%)

Hydrogen = 1 part (11.1%)

* Law of Multiple Proportions (Dalton)

When two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element are in a simple whole number ratio.

Example : Carbon and Oxygen
 $\text{CO} \rightarrow 12 \text{ g C combines with } 16 \text{ g O}$
 $\text{CO}_2 \rightarrow 12 \text{ g C combines with } 32 \text{ g O}$
 $16 : 32 = 1 : 2$ (simple whole number ratio)



1.6 Significant Figures

These are the digits in a measured value that are known with certainty plus the first uncertain digit.

Rules :

- ✓ All non-zero digits are significant.
- ✓ Zeros between non-zero digits are significant.
- ✓ Leading zeros are not significant.
- ✓ Trailing zeros are significant in a decimal number.
- ✓ Trailing zeros in whole numbers are not significant unless a decimal point is present.

Examples	No. of Significant Figures
25.67	4
0.00450	3
1200.	4
0.0203	3
1000	1

Remember!

Accuracy tells how close a measurement is to the true value.
Precision tells how close the measurements are to each other.

1.7 Accuracy and Precision

Accuracy → How close a measured value is to the true value.

Precision → How close repeated measurements are to each other.



Not Accurate
Not Precise



Precise
Not Accurate



Accurate
Not Precise



Accurate
Precise

Target Analogy

1.8 Errors in Measurement

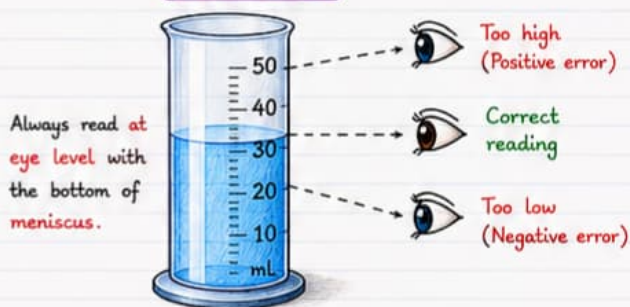
Errors are the difference between the measured value and the true value.

Type of Error	Description
Systematic Errors	- Arise due to faulty instrument or wrong method. - Affect accuracy.
Random Errors	- Arise due to unpredictable changes in conditions. - Affect precision.
Gross Errors	- Due to human mistakes or carelessness. - Can be minimized by repeating measurement.

Minimization of Errors

- ✓ Use of well-calibrated instruments.
- ✓ Take multiple readings and calculate average.
- ✓ Use proper and standardized methods.
- ✓ Avoid parallax error in reading.
- ✓ Maintain constant conditions.

Parallax Error



1.9 Dimensional Analysis

It is a mathematical method to check the correctness of a chemical equation or to derive relation between physical quantities.

Fundamental Dimensions

[M] - Mass [L] - Length [T] - Time [N] - Amount of substance

Examples of Dimensions

- Density (d) = Mass/Volume → $[d] = M L^{-3}$
- Speed (v) = Distance/Time → $[v] = L T^{-1}$
- Force (F) = Mass × Acceleration → $[F] = M L T^{-2}$

Uses

- ★ To check the correctness of equations.
- ★ To convert units.
- ★ To derive formulas.
- ★ To identify missing terms in equations.

Check the Equation

Example :

$$\text{Force} = \text{Mass} \times \text{Acceleration}$$

$$F = m \times a$$

$$\text{L.H.S.} \rightarrow [F] = M L T^{-2}$$

$$\begin{aligned} \text{R.H.S.} \rightarrow [m] \times [a] \\ = M \times L T^{-2} \\ = M L T^{-2} \end{aligned}$$

$$\text{L.H.S.} = \text{R.H.S.} \quad \checkmark$$

Equation is dimensionally correct.

1.10 Some Common Units

Physical Quantity	SI Unit	Symbol	Other Units	Symbol
Length	metre	m	centimetre, kilometre	cm, km
Mass	kilogram	kg	gram, milligram	g, mg
Time	second	s	minute, hour	min, h
Temperature	kelvin	K	degree Celsius	°C
Amount of substance	mole	mol	—	—
Electric current	ampere	A	milliampere	mA
Luminous intensity	candela	cd	—	—

Conversion of Units

- ★ $1 \text{ km} = 10^3 \text{ m}$
- ★ $1 \text{ cm} = 10^{-2} \text{ m}$
- ★ $1 \text{ g} = 10^{-3} \text{ kg}$
- ★ $1 \text{ mg} = 10^{-3} \text{ g}$
- ★ $1 \text{ L} = 10^{-3} \text{ m}^3$
- ★ $1 \text{ mL} = 10^{-6} \text{ m}^3$



Units allow us to compare and quantify things accurately.



Chemistry begins where numbers meet nature.

1.11 The Mole Concept and Molar Masses

Mole is the bridge between microscopic particles and macroscopic measurements.



The Mole Concept

- A mole is the amount of substance that contains as many elementary entities as there are atoms in exactly 12 g of carbon-12.
- 1 mole of any substance contains $N_A = 6.022 \times 10^{23}$ particles. (N_A = Avogadro constant)
- Mole helps us to count a very large number of particles in terms of grams.
- It is a counting unit just like dozen (12 items).

Examples (1 mole contains)



$$6.022 \times 10^{23} \text{ Helium atoms} = 4.00 \text{ g He}$$



$$6.022 \times 10^{23} \text{ Molecules of } O_2 = 32.00 \text{ g } O_2$$



$$6.022 \times 10^{23} \text{ Molecules of } H_2O = 18.00 \text{ g } H_2O$$



$$6.022 \times 10^{23} \text{ Formula units of NaCl} = 58.44 \text{ g NaCl}$$

Molar Mass

- The mass of 1 mole of a substance is called its **molar mass**.
- Its numerical value is equal to the relative atomic mass (for atoms) or relative molecular mass (for molecules) or formula mass (for formula units) in grams.
- Unit $\rightarrow \text{g mol}^{-1}$

How to Calculate Molar Mass?

Molar mass = Sum of atomic masses of all the atoms present in one formula unit of the substance.

Examples :

- (i) Molar mass of H_2O
 $= (2 \times 1) + 16 = 18 \text{ g mol}^{-1}$
- (ii) Molar mass of CO_2
 $= 12 + (2 \times 16) = 44 \text{ g mol}^{-1}$
- (iii) Molar mass of Na_2SO_4
 $= (2 \times 23) + 32 + (4 \times 16) = 142 \text{ g mol}^{-1}$

Key Point

$$\frac{\text{Mass (g)}}{n \text{ (mole)}} = \text{Molar mass (g mol}^{-1}\text{)}$$

$$n = \frac{\text{Mass}}{\text{Molar mass}}$$

$$\text{Mass} = n \times \text{Molar mass}$$

Mole Relationships

- $\text{Mass (g)} = n \times \text{Molar mass (g mol}^{-1}\text{)}$
- $n \text{ (mole)} = \frac{\text{Mass (g)}}{\text{Molar mass (g mol}^{-1}\text{)}}$
- Number of particles $= n \times N_A$

Example :

What is the mass of 0.250 mole of $CaCO_3$?

$$\text{Molar mass of } CaCO_3 = 40 + 12 + (3 \times 16) = 100 \text{ g mol}^{-1}$$

$$\text{Mass} = 0.250 \times 100 = 25.0 \text{ g}$$



1.12 Empirical and Molecular Formulas

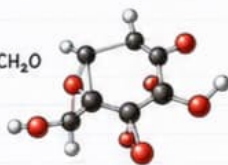
Empirical Formula

- The simplest whole number ratio of atoms present in a compound is called its empirical formula.
- It shows the simplest composition of the compound.
- It may or may not represent the actual number of atoms in the molecule.

Example :

Glucose ($C_6H_{12}O_6$) $\rightarrow$ Empirical formula $= CH_2O$

$$\begin{aligned} &= C : H : O \\ &= 6 : 12 : 6 \\ \text{Divide by 6} &\rightarrow 1 : 2 : 1 \\ &(\text{whole numbers}) \end{aligned}$$



Glucose ($C_6H_{12}O_6$)

Molecular Formula

- The actual number of atoms of each element present in one molecule of the compound is called its molecular formula.
- It is a whole number multiple of the empirical formula.

Example :

Glucose $\rightarrow$ Molecular formula $= C_6H_{12}O_6$

$$\begin{aligned} \text{Empirical formula} &= CH_2O \\ \text{Molecular formula} &= C_6H_{12}O_6 \\ \text{Multiple} &= 6 \\ (C_6H_{12}O_6 &= 6 \times CH_2O) \end{aligned}$$



Relation Between Them

$$\text{Molecular formula} = (\text{Empirical formula}) \times n$$

$$\text{Where, } n = \frac{\text{Molar mass of compound}}{\text{Empirical formula mass}}$$



How to Determine Molecular Formula?

- Find empirical formula of the compound from the % composition (or mass data).
- Calculate empirical formula mass (EFM).
- $n = \frac{\text{Molar mass of compound}}{\text{EFM}}$
- Molecular formula $= \text{Empirical formula} \times n$



Example : A compound contains 40.0% C, 6.7% H and 53.3% O. Molar mass of compound $= 180 \text{ g mol}^{-1}$. Find its molecular formula.

Step 1: Assume 100 g
 $C = 40.0 \text{ g} \rightarrow \frac{40.0}{12} = 3.33 \text{ mol}$
 $H = 6.7 \text{ g} \rightarrow \frac{6.7}{1} = 6.7 \text{ mol}$
 $O = 53.3 \text{ g} \rightarrow \frac{53.3}{16} = 3.33 \text{ mol}$
 Ratio $= 3.33 : 6.7 : 3.33$
 Divide by 3.33 $\rightarrow 1 : 2 : 1$
 Empirical formula $= CH_2O$

Step 2: Empirical formula mass (EFM)
 $= (12) + (2 \times 1) + (16)$
 $= 30 \text{ g mol}^{-1}$

$$\text{Step 3: } n = \frac{180}{30} = 6$$

Step 4: Molecular formula
 $= (CH_2O) \times 6$
 $= C_6H_{12}O_6$

Answer:
 $C_6H_{12}O_6$

1.13 Stoichiometry and Limiting Reagents



Stoichiometry

It is the calculation of quantitative relationships between reactants and products in a chemical reaction.

Steps in Stoichiometric Calculations

1. Balance the chemical equation.
2. Convert given quantities to moles.
3. Use mole ratio from balanced equation.
4. Convert to required quantity and unit.

Example :

In the reaction, $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
How many grams of water are formed from 4 g of H_2 ?

$$n(\text{H}_2) = \frac{4}{2} = 2 \text{ mol}$$

$$\text{Mole ratio } \text{H}_2 : \text{H}_2\text{O} = 2 : 2 = 1 : 1$$

$$\text{So, } n(\text{H}_2\text{O}) = 2 \text{ mol}$$

$$\text{Mass of } \text{H}_2\text{O} = 2 \times 18 = 36 \text{ g}$$

Limiting Reagents

- * The reactant that is completely consumed first in a chemical reaction is called the limiting reagent.
- * It determines the maximum amount of product that can be formed.
- * Other reactants are left in excess.

How to Find Limiting Reagent?

1. Convert all given reactants to moles.
2. Divide by their stoichiometric (mole) coefficients.
3. The smallest value gives the limiting reagent.

Example :

For the reaction: $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$

Given: 28 g N_2 and 10 g H_2

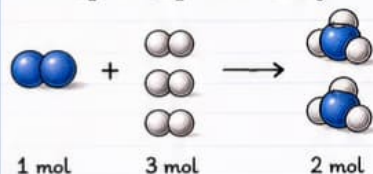
Find the limiting reagent.

Reactant	Given (g)	Moles (n)	Coefficient	Coefficient
N_2	28	$\frac{28}{28} = 1$	1	$\frac{1}{1} = 1$
H_2	10	$\frac{10}{2} = 5$	3	$\frac{5}{3} = 1.67$

Smallest value = 1 (for N_2)

$\therefore \text{N}_2$ is the limiting reagent.

Visual Understanding



Here H_2 is in excess.
Some H_2 will be left unreacted.



Key Points

- Limiting reagent limits the amount of product.
- It gets used up completely.
- Excess reagent remains after reaction.
- Important in industrial and laboratory calculations.

1.14 Methods to Express Concentration of Solutions

Concentration tells us how much solute is present in a given amount of solution or solvent.



1. Mass Percentage (w/w%)

Mass of solute in grams per 100 g of solution.

$$\text{Mass \%} = \frac{\text{Mass of solute (g)}}{\text{Mass of solution (g)}} \times 100$$

Example : 10 g of NaCl is dissolved in 90 g of water.

$$\text{Mass \%} = \frac{10}{100} \times 100 = 10\%$$



2. Volume Percentage (v/v%)

Volume of solute in mL per 100 mL of solution.

$$\text{Volume \%} = \frac{\text{Volume of solute (mL)}}{\text{Volume of solution (mL)}} \times 100$$

Example : 20 mL of ethanol in 80 mL of water.

$$\text{Volume \%} = \frac{20}{100} \times 100 = 20\%$$



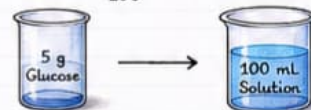
3. Mass by Volume Percentage (w/v%)

Mass of solute in grams present in 100 mL of solution.

$$\text{w/v \%} = \frac{\text{Mass of solute (g)}}{\text{Volume of solution (mL)}} \times 100$$

Example : 5 g of glucose in 100 mL of solution.

$$\text{w/v \%} = \frac{5}{100} \times 100 = 5\%$$



4. Molarity (M)

Moles of solute per litre of solution.

$$M = \frac{\text{Moles of solute (mol)}}{\text{Volume of solution (L)}}$$

Example : 0.5 mol NaCl in 1 L solution.

$$M = \frac{0.5}{1} = 0.5 \text{ M}$$



5. Molality (m)

Moles of solute per kilogram of solvent.

$$m = \frac{\text{Moles of solute (mol)}}{\text{Mass of solvent (kg)}}$$

Example : 2 mol urea in 1.5 kg water.

$$m = \frac{2}{1.5} = 1.33 \text{ m}$$



6. Normality (N)

Number of gram-equivalents of solute per litre of solution.

$$N = \frac{\text{Gram-equivalents of solute}}{\text{Volume of solution (L)}}$$

Gram-equivalents = $\frac{\text{Mass of solute}}{\text{Equivalent mass}}$

Example : 1 eq. NaOH in 1 L solution.

$$N = \frac{1}{1} = 1 \text{ N}$$



Quick Comparison

Method	Based On
Mass % (w/w%)	Mass of solute in 100 g solution
Volume % (v/v%)	Volume of solute in 100 mL solution
Mass by Volume % (w/v%)	Mass of solute in 100 mL solution
Molarity (M)	Moles of solute in 1 L solution
Molality (m)	Moles of solute in 1 kg solvent
Normality (N)	Gram-equivalents in 1 L solution

Remember!

Different expressions of concentration are used in different situations. Choose the right one based on the data available!



Why Concentration Matters?

- ✓ Used in preparing medicines and solutions.
- ✓ Important in chemical industries and labs.
- ✓ Helps in stoichiometric calculations.
- ✓ Affects reaction rate and equilibrium.

